THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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ANSWER KEY & MARKING SCHEME SET A

Chapter 12: **Surface Areas and Volumes** | Class 10 Mathematics | Maximum Marks: 40

SECTION — A (Objective Type) 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** $3\pi r^2$	Curved $2\pi r^2$ + base πr^2
2	**(a)** $\frac{1}{3}\pi r^2 h + \frac{2}{3}\pi r^3$	Sum of the two volumes
3	**(a)** 250 cm²	$2(50 + 25 + 50)$
4	**(a)** 3 : 1	$\pi r^2 h : \frac{1}{3}\pi r^2 h$
5	**(b)** Incorrect — the surfaces in contact are not counted	Joined faces become hidden
6	**(a)** 19404 cm³	$\frac{2}{3} \times \frac{22}{7} \times 9261$
7	**(a)** Student A is correct	No flat base is exposed
8	**(b)** $2\pi r^2$	Two new circular faces
9	**(a)** 1540 cm³	$\frac{22}{7} \times 49 \times 10$
10	**(a)** 25 cm	$\sqrt{576 + 49}$

SECTION — B (Very Short Answer) — Answers 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$1540 + \frac{2}{3} \times \frac{22}{7} \times 343 = 1540 + 718.67 = 2258.67$ cm³	1+1
12	$2\pi rh + \pi rl = \frac{22}{7} \times 7(6 + 10) = 352$ m²	1+1
13	$192 = 2 \times 96$ counts the two hidden faces. Correct: $2(32 + 16 + 32) = 160$ cm². Claim is false.	1+1
14	Cone $= 77$, hemisphere $= 89.83 \Rightarrow 166.83$ cm³	1+1

SECTION — C (Short Answer) — Answers 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Apparent $= 3.14 \times 6.25 \times 10 = 196.25$ cm³; hemisphere $= \frac{2}{3} \times 3.14 \times 15.625 = 32.71$; actual $= 163.54$ cm³	1+1+1
16	$525 - 4 \times \frac{1}{3} \times \frac{22}{7} \times 0.25 \times 1.4 = 525 - 1.47 = 523.53$ cm³	1+1+1
17	Diameter 7 cm. SA $= 6 \times 49 - \pi r^2 + 2\pi r^2 = 294 + 38.5 = 332.5$ cm²	1+2
	OR: $6l^2 - \frac{\pi l^2}{4} + \frac{\pi l^2}{2} = \frac{l^2}{4}(24 + \pi)$	1+1+1
18	$\frac{1}{3} \times \frac{22}{7} \times 12.25 \times 12 + \frac{2}{3} \times \frac{22}{7} \times 42.875 = 154 + 89.83 = 243.83$ cm³	1+1+1
	OR: $l = 5$: $\pi rl = 47.1$, $2\pi rh = 150.72$, base $= 28.26 \Rightarrow 226.08$ cm²	1+1+1

SECTION — D (Long Answer) — Answers 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $2\pi r(h + r) = 2 \times \frac{22}{7} \times 30 \times 175 = 33000$ cm² $= 3.3$ m² (ii) $\frac{2}{3} \times \frac{22}{7} \times 27000 \approx 56571$ cm³ ≈ 56.57 L	3+2
	OR: Cylinder length 72: $2\pi rh \approx 8146.29$; sphere $4\pi r^2 \approx 4073.14$; total ≈ 12219.43 cm²; cost $\approx$ ₹855.36	2+2+1
20	(i) 141.3 cm³ (ii) 56.52 cm³ (iii) 197.82 cm³ (iv) 9891 cm³ ≈ 9.89 L	1+1+2+1
	OR: (i) 25 cm (ii) $550 + 308 = 858$ cm² (iii) ₹429 (iv) $1232 + 718.67 = 1950.67$ cm³	1+2+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET B

Chapter 12: **Surface Areas and Volumes** | Class 10 Mathematics | Maximum Marks: 40

SECTION — A (Objective Type) — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 220 cm²	$\frac{22}{7} \times 7 \times 10$
2	**(a)** $\frac{4}{3}\pi r^3$	It is a sphere
3	**(a)** $2\pi r(r+h)$	Curved + two ends
4	**(a)** 66 cm³	$\frac{1}{3} \times \frac{22}{7} \times 9 \times 7$
5	**(a)** Correct	Volumes add when parts do not overlap
6	**(a)** Student A is correct	$440 + 154 + 308 = 902$; B counted the hidden top face
7	**(b)** 2 : 1	$\frac{1}{3}\pi r^2 h = \frac{2}{3}\pi r^3 \Rightarrow h = 2r$
8	**(a)** 179.67 cm³	$r = 3.5$: $\frac{4}{3} \times \frac{22}{7} \times 42.875$
9	**(a)** 308 cm²	$2 \times \frac{22}{7} \times 49$
10	**(b)** the base of the hemisphere and the top of the cylinder	Faces in contact are hidden

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$l = 10$: $\pi r l = 188.4$; $2\pi r^2 = 226.08 \Rightarrow 414.48$ cm²	1+1
12	$\frac{880}{7} + \frac{88}{7} = \frac{968}{7} \approx 138.29$ cm³	1+1
13	Volume $= 54$ cm³ (claim false). Cuboid $6 \times 3 \times 3$: SA $= 2(18+9+18) = 90$ cm²	1+1
14	$r = 3.5$, cylinder 14 mm: $2\pi r h = 308$, $4\pi r^2 = 154 \Rightarrow 462$ mm²	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	SA $= 220 + 154 = 374$ cm²; volume $= 385 + 179.67 = 564.67$ cm³	1½+1½
16	Cone $= 18.48$, hemisphere $= 19.404 \Rightarrow \approx 37.88$ cm³	1+1+1
17	$l = 2.5$: cylinder CSA $= 10.56$, cone CSA $= 5.5$, base $= 1.54 \Rightarrow 17.6 \approx 18$ cm²	1+1+1
	OR: $198 - \frac{264}{7} = \frac{1122}{7} \approx 160.29$ cm³	1+1+1
18	$2\pi r^2 + 2\pi r h = 2 \times \frac{22}{7} \times 10.5(10.5+14) = 1617$ cm²	1+1+1
	OR: $264 + 55.44 + 110.88 = 430.32$ cm²	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Height $2r \Rightarrow$ cylinder height r. $\pi r^3 + \frac{2}{3}\pi r^3 = \frac{5}{3}\pi r^3 = \frac{880}{21} \Rightarrow r^3 = 8$, $r = 2 \Rightarrow$ height $= 4$ m	2+2+1
	OR: Cylinder 2 m. (i) $12.32 + 5.749 = 18.069$ m³ ≈ 18069 L (ii) $2\pi r h + 2\pi r^2 = 17.6 + 12.32 = 29.92$ m²	3+2
20	(i) 5.775 cm³ (ii) 0.154 cm³ (iii) 5.929 cm³ (iv) $l = 1.25$ cm; $\pi r l = 1.375$ cm²	1+1+1+2
	OR: (i) $\pi r(2h+l) = \frac{22}{7} \times 52.5 \times 59 = 9735$ m² (ii) ₹116820	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.